

FIZ1951-MUHENDİSLER İÇİN YARIİLETKEN FİZİĞİ FORMÜL KAĞIDI

Özellik / Malzeme	Silisyum (Si)	Germanyum (Ge)	Galyum Arsenik (GaAs)
Dielektrik sabiti (ϵ)	11.7	16.0	13.1
E_g (eV) (@300K)	1.12	0.67	1.43
m_n^*/m_0	1.08	0.55	0.067
m_h^*/m_0	0.99	0.37	0.47
N_C (cm ⁻³) (@300K)	2.8×10^{19}	1.0×10^{19}	4.3×10^{17}
N_V (cm ⁻³) (@300K)	2.5×10^{19}	5.4×10^{18}	8.1×10^{18}
μ_n (cm ² /V.s) (@300K)	1350	3800	8500
μ_p (cm ² /V.s) (@300K)	480	1800	450

$$\begin{aligned} \epsilon_0 &= 8.85 \times 10^{-12} \frac{F}{m} & q &= e = 1.6 \times 10^{-19} C \\ h &= 6.63 \times 10^{-34} J.s & m_0 &= 9.11 \times 10^{-31} kg \\ k_B &= 1.38 \times 10^{-23} J/K \end{aligned}$$

Elektron: $m_n = m_e$

Boşluk: Hol: Deşik: $m_h = m_b$

$$n_i = CT^{3/2} e^{-E_g/2k_B T}$$

$$\sigma_i = n_i e (\mu_p + \mu_n)$$

$$n_i \cdot p_i = n_i^2 = p_i^2$$

$$n_i = p_i$$

$$C = 2 \left(\frac{2\pi k_B \sqrt{m_e^* m_h^*}}{h^2} \right)$$

$$E_g = E_C - E_V$$

$$f_n(E, T) = \frac{1}{1 + e^{\frac{E - E_F}{k_B T}}}$$

$$f_p(E, T) = \frac{1}{1 + e^{\frac{E_F - E}{k_B T}}}$$

$$1 \geq f_n(E, T) \geq 0$$

$$N_{CB}(E) = \frac{1}{2\pi^2 \hbar^3} (2m_e^*)^{3/2} (E - E_C)^{1/2}$$

$$N_{VB}(E) = \frac{1}{2\pi^2 \hbar^3} (2m_h^*)^{3/2} (E_V - E)^{1/2}$$

$$N_C = 2 \left(\frac{2\pi m_e^* k_B T}{h^2} \right)^{3/2}$$

$$n_i = N_C \exp \left(-\frac{E_C - E_F}{k_B T} \right)$$

$$N_V = 2 \left(\frac{2\pi m_h^* k_B T}{h^2} \right)^{3/2}$$

$$p_i = N_V \exp \left(-\frac{E_F - E_V}{k_B T} \right)$$

$$n_i = \sqrt{N_C N_V} \exp \left(-\frac{E_g}{2k_B T} \right)$$

$$\frac{e}{k_B T} = \frac{\mu_p}{D_p} = \frac{\mu_n}{D_n}$$

$$J_p = -e D_p \frac{dp}{dx} \quad J_n = e D_n \frac{dn}{dx}$$

$$D_n = \mu_n \frac{k_B T}{q} \quad D_p = \mu_p \frac{k_B T}{q}$$

$$\sigma_p = pq\mu_p \quad \sigma_n = nq\mu_n$$

$$E_{Fi} = E_V + \frac{1}{2} E_g + \frac{1}{2} k_B T \ln \frac{N_V}{N_C}$$

$$E_{Fi} = E_V + \frac{1}{2} E_g + \frac{3}{4} k_B T \ln \frac{m_h^*}{m_e^*}$$

$$\vartheta_t = \sqrt{\frac{3k_B T}{m^*}}$$

$$\sigma = \frac{ne^2 \tau}{m_e^*}$$

$$J_{Top} = \left(en\mu_n E + e D_n \frac{dn}{dx} \right) + \left(ep\mu_p E - e D_p \frac{dp}{dx} \right)$$

$$\begin{aligned} N_A \text{ ve } N_D \gg n_i &\Rightarrow n \text{ tipi} \\ N_A \text{ ve } N_D \gg n_i &\Rightarrow p \text{ tipi} \end{aligned}$$

$$n = n_i + N_D \quad N_D \gg n_i \rightarrow n \approx N_D$$

$$p = \frac{n_i^2}{N_D} \ll N_D$$

$$\sigma_n \approx N_D e \mu_n$$

$$p = p_i + N_A \quad N_A \gg p_i \rightarrow p \approx N_A$$

$$p = \frac{n_i^2}{N_A} \ll N_A$$

$$\sigma_p \approx N_A e \mu_p$$

$$E_{Fn} = E_C + k_B T \ln \left(\frac{n}{N_C} \right)$$

$$n = N_C \exp \left(-\frac{E_C - E_{Fn}}{k_B T} \right)$$

$$E_{Fp} = E_V - k_B T \ln \left(\frac{p}{N_V} \right)$$

$$p = N_V \exp \left(-\frac{E_{Fp} - E_V}{k_B T} \right)$$

$$n = \frac{(N_D - N_A)}{2} + \sqrt{\left(\frac{N_D - N_A}{2} \right)^2 + n_i^2}$$

$$\frac{1}{\mu_n^T} = \frac{1}{\mu_n^L} + \frac{1}{\mu_n^I}$$

$$\frac{qpV_H}{B_z} = \frac{I_x}{t}$$

$$E_{Hall} = v_s B, \quad v_s = \frac{I}{nqA}, \quad A = tW$$

$$V_{Hall} = R_{Hall} \frac{IB}{t}, \quad R_{Hall} = \frac{1}{nq}$$

$$q(\vec{\partial} \times \vec{B}) = q\vec{E}_{Hall}$$

$$qv_s B = qE_{Hall}$$

$$\vec{x} = \vec{v}_s \Delta t$$

$$I = \frac{\Delta Q}{\Delta t}$$

$$I = nqA\vartheta_s$$

$$J = \frac{I}{A}$$

$$\vec{J} = nq\vec{\vartheta}_s$$

$$V = IR$$

$$R = \rho \frac{l}{A}$$

$$\vec{v}_s = \mu \vec{E} = \frac{q\tau}{m} \vec{E}$$

$$\vec{J} = nq\mu \vec{E}$$

$$\sigma = nq\mu = ne\mu$$

$$q = e$$

$$\mu_n = \frac{e\tau_n}{m_n^*}$$

$$\mu_p = \frac{e\tau_p}{m_p^*}$$

$$\vec{J} = \sigma \vec{E} \quad \vec{E} = \frac{V}{l}$$

$$\rho = \frac{1}{ne\mu}$$

$$l = u_{ort} \tau$$

$$u_{th} = l / \tau$$

$$\sigma = 1/\rho \rightarrow S = 1/R$$

$$\vec{F}_e = q\vec{E} = m_o \vec{a}$$

$$\vec{F}_e = q\vec{E} + q\vec{E}_{Kr} = m^* \vec{a}$$

$$E_f = \frac{h^2}{4m} \cdot \left(\frac{3}{\pi} \cdot n \right)^{2/3}$$

$$E_f = \frac{1}{2} \cdot m \cdot v_f^2$$

$$F_n = -q_{th} \cdot l \cdot \frac{dn}{dx}$$

↓

elektronun net
akış oranı

$$l \cdot v_{th} = D_n / D_p$$

$$v_f = \left(\frac{2E_f}{m_e} \right)^{1/2}$$

$$J = eV$$

↓
q

$$\frac{M}{V} = \rho$$

$$V = \frac{Na}{\rho} \cdot \frac{1}{6.02 \cdot 10^{23}}$$

hiz n sayı
vitr.

$T=0$ u'le Fermi Enerji Seviyesi
için $\Rightarrow$ kimyasal potansiyel denir.